

Viyom Shukla

LinkedIn: viyom-shukla

GitHub: github.com/viyomshukla

TUF+: 350+ DSA problems | LeetCode: 200+ problems

Email: Viyomshukla21@gmail.com

Mobile: +91-6397332481

Portfolio: viyom-potfolio.vercel.app

Location: Noida, India

EDUCATION

- Jaypee Institute of Information Technology** Noida, India
 - Integrated B.Tech + M.Tech in Computer Science Engineering; CGPA: 7.62/10 2022 – 2027

TECHNICAL SKILLS

Programming Languages: C, C++ (Proficient), JavaScript, Python, HTML, CSS, SQL

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS

AI & Retrieval: Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), Hybrid Search (Embeddings), LLM APIs, GANs

Tools & Technologies: Git, GitHub, MongoDB, MySQL, API, VS Code, Firebase, Cloudinary, Postman, Slack

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Machine Learning

EXPERIENCE

- Distance Connect** Remote
 - Full Stack Web Intern Aug 2025 – Oct 2025
 - Conversational AI Service Integration:** Engineered and deployed an LLM-backed chatbot to automate student onboarding and query resolution, handling the highest-volume support paths without manual intervention.
 - Third-Party API Integration:** Built a real-time mentorship flow by integrating the Google Meet API with OAuth-based authorization, enabling secure 1:1 virtual sessions to be scheduled and launched directly from the platform.
 - Platform UI Revamp:** Led frontend development for an early-stage EdTech platform connecting students with IIT and AIIMS mentors, rebuilding the interface to improve student engagement.
- Student Placement Committee, IIIT Noida** Noida, India
 - Coordinator Aug 2024 – Jan 2025
 - Corporate Outreach:** Coordinated with the placement head and cross-functional student teams to source companies actively hiring for campus roles, compiling job requirements and eligibility criteria to prioritize outreach efforts.

PROJECTS

- Thumbblify — AI-Powered Thumbnail Generation & Editing Platform (React.js, Node.js, MongoDB, REST APIs, Cloudinary)** Live
 - Fault-Tolerant Multi-Provider Routing:** Built a provider-agnostic routing layer over Replicate APIs with automatic failover across multiple LLM backends, eliminating single-provider downtime and sustaining 100+ users generating 2K/4K thumbnails without service interruption.
 - Secure Payment Infrastructure:** Eliminated transaction-tampering risk in the credit-based subscription model by implementing server-side Razorpay webhook signature verification, achieving 100% payment integrity across all completed transactions.
 - Multi-Channel HTTP Gateway:** Reduced generation friction for mobile users by building a Twilio-integrated WhatsApp bot alongside an in-app chatbot, enabling prompt-to-thumbnail generation without requiring users to open the web dashboard.
- Hiring Copilot — RAG Pipeline for Resume-JD Matching (Python, FastAPI, pdfplumber, Pydantic, LLM APIs, LangGraph, ChromaDB, SQLite, Next.js)** Source Code(In Progress)
 - Document Ingestion & Parsing Pipeline:** Built a Python service that parses bulk PDF/DOCX resumes via pdfplumber and python-docx into normalized text, with a custom chunking layer validated by an evaluation harness that measures retrieval quality across chunking configurations.
 - Hybrid Retrieval & Reranking:** Designed a two-stage retrieval layer fusing BM25 keyword search with dense vector search over ChromaDB via Reciprocal Rank Fusion, then reranking with a sentence-transformers cross-encoder to return cited resume evidence per job requirement.
 - Structured Extraction & Agent Orchestration:** Implementing schema-enforced LLM extraction with Pydantic to convert unstructured resumes into typed records in SQL storage, orchestrated as a LangGraph agent workflow with a ragas-driven evaluation loop.
- NextTrip — Booking & Management Platform (Backend: Node.js, MongoDB, REST APIs)** Live
 - Secure Access & Transactions:** Engineered role-based access control via Google OAuth 2.0 and JWT for admin/traveler roles, securing 100% of API routes, and automated Razorpay payments with server-side verification, reducing manual reconciliation to 0.
 - KYC-Compliant Identity Module:** Built an Aadhaar document-upload flow with admin review and manual approval, on a normalized 6-collection MongoDB schema with indexed queries backing an admin review dashboard.
 - Admin Control & Data Layer:** Built an admin dashboard for trip listings, promotional imagery, and booking analytics on a normalized 6-collection MongoDB schema with indexed queries.

RESEARCH & PUBLICATIONS

- Enhancing Network Intrusion Detection Using Conditional GANs for Imbalanced Data Classification** IEEE Xplore
- Enhancing Network Intrusion Detection Using Quantum GANs for Imbalanced Data Classification** IEEE — Under Review

ACHIEVEMENTS

- International Conference Speaker:** Presented published research at the 17th IEEE IC3-2025, IIIT Noida. *Certificate*
- Competitive Programming:** Solved 550+ DSA and 50+ SQL problems across TUF+, LeetCode, and GeeksforGeeks, focusing on optimization, time complexity, and algorithm design.
- Smart India Hackathon (SIH '23):** Selected among the top teams across India in the Tourism domain.
- National Hackathon Competitor:** Competed in the Amazon ML Challenge and Adobe India Hackathon '25.